

CNC-X100 Vertical Machining Center Installation Guide

Product: CNC Machine | Model: CNC-X100 | Document Type: INSTALLATION_GUIDE

1. Site Preparation & Environmental Requirements

Proper site preparation ensures long-term machining positioning accuracy, operator safety, and structural machine stability.

- Foundation Requirement: Heavy-duty reinforced concrete pad with minimum thickness of 200 mm (8 inches).
- Floor Load Rating: Minimum load-bearing capacity of 3,500 kg/m².
- Service Access Clearance: Minimum 1.0 meter (3.3 feet) open clearance around machine rear and sides for maintenance access.
- Operating Ambient Temperature: +15°C to +35°C (+59°F to +95°F). Optimal operating temperature: 20°C –2°C.
- Relative Humidity: 30% to 80% non-condensing.
- Vibration Isolation: Keep machine at least 15 meters away from stamping presses, forging hammers, or heavy overhead cranes.

2. Uncrating, Rigging & Machine Leveling

Unpacking and hoisting procedures must be performed by certified industrial rigging personnel:

1. Use an overhead crane (minimum 5-ton capacity) or 5-ton forklift to position machine over concrete pad.
2. Install six heavy-duty steel leveling pads beneath base casting adjustment bolts.
3. Place precision spirit level (0.02 mm/m sensitivity) on ground table surface.
4. Adjust leveling bolts until table is level within 0.02 mm/m across both X and Y directions.

3. Electrical Installation & Grounding Requirements

Electrical installation must comply with national electrical codes and industrial facility safety standards:

- Input Power Supply: 3-Phase AC 400V –10%, 50/60 Hz.
- Total Connected Electrical Load: 35 kVA. Full load current rating: 48 Amps.
- Main Power Cable: 4-core copper cable, minimum cross-section 16 mm².
- Circuit Breaker Rating: Recommended 63 Amp 3-pole circuit breaker with residual current circuit breaker (RCCB).
- Control Voltage: 24V DC regulated internal power supply.
- Earth Grounding: Dedicated ground rod with ground resistance less than 5

Ohms. Do not share ground line with arc welders.

4. Compressed Air & Coolant System Setup

Connect facility compressed air line to pneumatic regulator FRL unit at rear of machine:

- Pneumatic Operating Pressure Range: 6.0 bar to 8.0 bar (87 psi to 116 psi). Recommended nominal pressure: 6.5 bar.
- Air Flow Rate: Minimum 250 Liters/minute continuous clean flow.
- Air Quality & Filtration: 5-micron filter with automatic water condensate drain.
- Air Pressure Warning: Pressure below 6.0 bar will trigger pneumatic interlock fault and prevent tool unclamping cycles.
- Coolant Tank Setup: Fill 200-Liter tank with 6% water-soluble synthetic coolant emulsion to 80% fill mark.

5. Initial Setup & Post-Installation Commissioning

Before applying axis drive power, complete initial calibration checks:

- Remove red shipping transit locking brackets from Z-axis ball screw and tool changer arm.
- Fill automatic central lubrication reservoir with ISO VG 68 slide-way oil.
- Perform spindle runout check using dial test indicator (maximum allowable runout 0.003 mm).
- Execute laser interferometer pitch error compensation calibration routine.

WARNING / SAFETY PRECAUTION:

Verify correct 3-phase phase rotation before turning on spindle. Incorrect phase rotation will reverse coolant pump and chip conveyor motors.